

Numerical and Experimental Study on Mitigation of Storm Geysers in Edmonton, Alberta, Canada

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Abstract: Geysers can occur in storm sewer systems during heavy storm events, leading to public safety concerns. In this study, a physical model as well as a transient three-dimensional computational fluid dynamic model were used to investigate potential mitigation methods. Two types of geysers, caused by the rapid filling and release of air pockets, were simulated numerically and validated with physical measurements. For geyser mitigation, benching, water recirculation chambers, and orifice plates were proposed and assessed numerically, and the effect of an inflow control method was also studied. The numerically tested geyser mitigation methods showed positive effects on reducing the amount of water erupting out of the manhole, but they also increased the amount of air transported downstream, which could cause issues further downstream. Water recirculation chambers mitigated the water erupting out of manholes without increasing the pressure in the pipes significantly. The strategy of sealing the riser was able to completely mitigate geyser events, but it also generated higher pressure in the upstream pipe and transported a higher amount of air downstream. It was found that using orifice plates to mitigate the second type geyser can generate a water hammer-like pressure pattern in the riser, with a peak pressure reaching about seven times the driving pressure when the orifice diameter is 0.2 times the riser diameter. With the inflow control method, the physically measured pressure variation was reduced, and the onset of geysers was visually reduced. DOI: 10.1061/(ASCE)HY.1943-7900.0001684. © 2019 American Society of Civil Engineers.

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Introduction

Geyser events in storm sewer systems are manifested as explosive ejections of an air/water mixture from manholes during heavy rain-storm events. They can generate public safety concerns, such as manhole covers blowing off, local flooding, and even system failure (Li and McCorquodale 1999; Shao 2013; Huang et al. 2018). In Edmonton, Alberta, Canada, several geyser events have been witnessed at one location (Liu et al. 2020). The schematic of this specific manhole is shown in Fig. 1. The new 3,500-mm tunnel ties into the existing 3,000-mm tunnel with a chamber that is 8 m in height and 5.8 m in diameter. The flow from the new tunnel turns 90° and

merges with the existing tunnel after a drop of approximately 4 m. A mitigation method is urgently needed for the site in order to prevent geysers, due to the sensitivity of this location and public safety concerns.

Triggers for geyser events can be categorized into two types. First, inertial oscillations may lead to water eruptions, such as geysers. A pipe-filling bore, upon reaching a drainage structure, may also trigger a geyser through an inertial oscillation (Guo and Song 1990; Zhou et al. 2002; Lewis 2011; Zhou et al. 2011; Li and Zhu 2018; Li et al. 2018). In this type of trigger, air is pushed out of the manhole. When the wave front propagates and reaches the manhole, a pressure peak is observed; the peak pressure can reach 15 times the driving pressure (Zhou et al. 2002) and cause significant damage to sewer infrastructure. Second, when air pockets in horizontal pipes reach a vertical shaft (referred to as *risers* in this study) and escape through the riser, they may trigger geysers (Vasconcelos and Wright 2009; Wright et al. 2008; Liu et al. 2020). In this type of geyser, an air pocket trapped in a sewer pipe moves toward a manhole. Once the air pocket reaches the manhole shaft, the upward motion of the air pocket tends to push the water in the shaft toward the manhole cover, causing a geyser. Choi et al. (2014, 2016), Cong et al. (2017), Chan et al. (2018), Wang and Vasconcelos (2018), and Muller et al. (2017) studied geysers triggered by the release of air pockets and found that computational fluid dynamics (CFD) can be used to study geyser events.

Extensive studies on geysers in storm sewer systems have indicated that the release of air pockets trapped in pipes triggers geyser events. Vasconcelos and Wright (2011) studied the air/water mixture flowing out of a riser connected directly to a horizontal pipe. The present study aimed to represent geyser episodes in a specific system, and the geometry was scale-reduced to represent this in detail. In the sewer system analyzed in the present study, there is a chamber beneath the riser, and the upstream and downstream pipes have different diameters and invert elevations. In prototype

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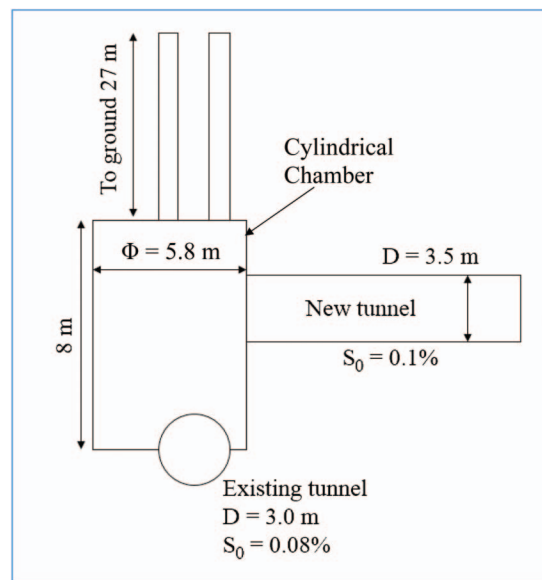


Fig. 1. (Color) Schematic of the problematic manhole.

storm sewer systems, the diameters of manhole covers are typically smaller than those of the manhole shafts, and therefore, it is more realistic to study geyser events using a riser with a smaller opening at the top end. Also, the displacement of manhole cover is of interest, because it proposes public safety issues.

Wright et al. (2009) proposed two conceptual methods for mitigating geysers. One was to add two 90° bends to the riser in order to create a near-horizontal section with the intent of separating the air phase from the liquid phase; the other was to expand the diameter of the top part of the shaft. Lewis (2011) physically modeled these two mitigation methods in the laboratory and found that the method using the additional horizontal pipe was not effective in preventing geyser events; the diameter expansion method was successful in reducing the strength of geyser but only in certain conditions. Wang and Vasconcelos (2017) examined possible mitigation methods by lowering the rim of the vertical shaft under the water in order to avoid the release of air pockets. Huang et al. (2018) proposed using an orifice plate to reduce the pressure in the horizontal pipe and the height of geysers; this method was used in the present study. The present paper aimed to comprehensively study possible mitigation methods for geyser events based on the Edmonton case. For a more general point of view, different potential geyser mitigation methods were tested related to two different geyser mechanisms; it has been confirmed that these mechanisms can trigger severe geysers in the Edmonton case, as has been reported in Liu et al. (2020).

There are two major differences between the current study and previous studies reported in the literature: (1) the existence of a chamber and the elevation change between the upstream and downstream pipes; and (2) a sudden change in the inflow rate. In Muller et al. (2017), Huang et al. (2018), and Cong et al. (2017), there was no chamber connecting the riser and the horizontal pipe. The air pocket that pushes the water column in the riser upward can also travel directly downstream, which may reduce the strength of a geyser event. In the current study, the existence of the chamber and the misalignment of the upstream and downstream pipes made the riser the only exit for the air. The entrapped air pocket had less chance of being transported further downstream. In Vasconcelos and Wright (2011), Cong et al. (2017), and Chan et al. (2018), the air pocket was predefined with no change in the inflow rate. In reality, the

hydrograph of a storm event may induce changes in inflow rate, and it is the flow rate variation that drives an entrapped air pocket to a manhole, where a geyser occurs. Therefore, the current study focuses more on prototype connections and can reflect the actual air/water interaction in prototype structures.

Benching and water recirculation chambers (WRCs) are alternative mitigation methods newly considered in the literature. Benching involves installing a pipe that changes diameter in the chamber that connects upstream and downstream, a conical expansion in this study. The top of the benching has multiple openings for releasing air pockets trapped in the upstream pipe. The purpose of the benching is to guide increased flow smoothly from the upstream pipe to the downstream pipe and reduce the turbulent mixing of air and water in the chamber. A WRC is a chamber located beside the riser that is connected to the top of the riser; its purpose is to collect the water that erupts out of a manhole during a geyser event. The top of the riser is connected to the chamber using a 90° bend, and the bottom of the chamber is connected to the riser to ensure that the water collected in the chamber can flow back into the storm sewer system. During rapid inflow conditions, the air/water mixture generated by a geyser event jetting out of a problematic manhole is redirected toward the WRC. The air/water mixture is separated in the WRC; the air is released to the atmosphere, and the water flows back to the riser.

Currently, understanding of geyser mitigation methods is still limited. To better assess potential geyser mitigation solutions, this study was conducted based on the aforementioned problematic manhole. A three-dimensional (3D) transient computational fluid dynamics model was developed. Three geyser mitigation methods, including a water recirculation chamber, benching, and an orifice plate, were employed to mitigate geysers triggered by a surge front while the downstream pipe was near full. Orifice plates with different opening sizes were tested for geysers triggered by an entrapped air pocket with water in the riser. An inflow control method was also tested. The purpose of this study is to provide municipalities with feasible mitigation methods for addressing geyser issues in the storm sewer system in Edmonton; some of these strategies could potentially be useful in other systems.

Methodology

The general methodology of this study involved building a lab-scale physical model that represented the prototype system and a numerical model that had the same dimensions as the physical model. The measurements included the pressure variations during geyser events and the amount of water flowing out of the riser. Water/air interactions in the chamber and riser were also investigated. A schematic of the physical model is shown in Fig. 2(a). It was designed based on the aforementioned prototype connection in Edmonton, Alberta, Canada. The upstream and downstream pipes had diameters of 0.20 and 0.28 m, respectively, and lengths of 6.00 and 5.95 m, respectively. The slope of the upstream pipe was 1:100, and the downstream pipe was horizontal. The dimensions of the chamber beneath the riser were $0.3 \times 0.3 \times 0.45$ m, and the crown of the upstream pipe had an elevation change relative to the top of the chamber. The diameter of the riser was 0.06 m, and its length was 1.22 m; it was roughly a 1:20 scale model.

A circular overflow weir was installed in the downstream tank to control the initial water level in the downstream pipe at the crown of the downstream pipe. A removable tailgate was built between the downstream pipe and the downstream tank in order to adjust the flow capacity of the downstream pipe and control the initial water level in the chamber and riser.

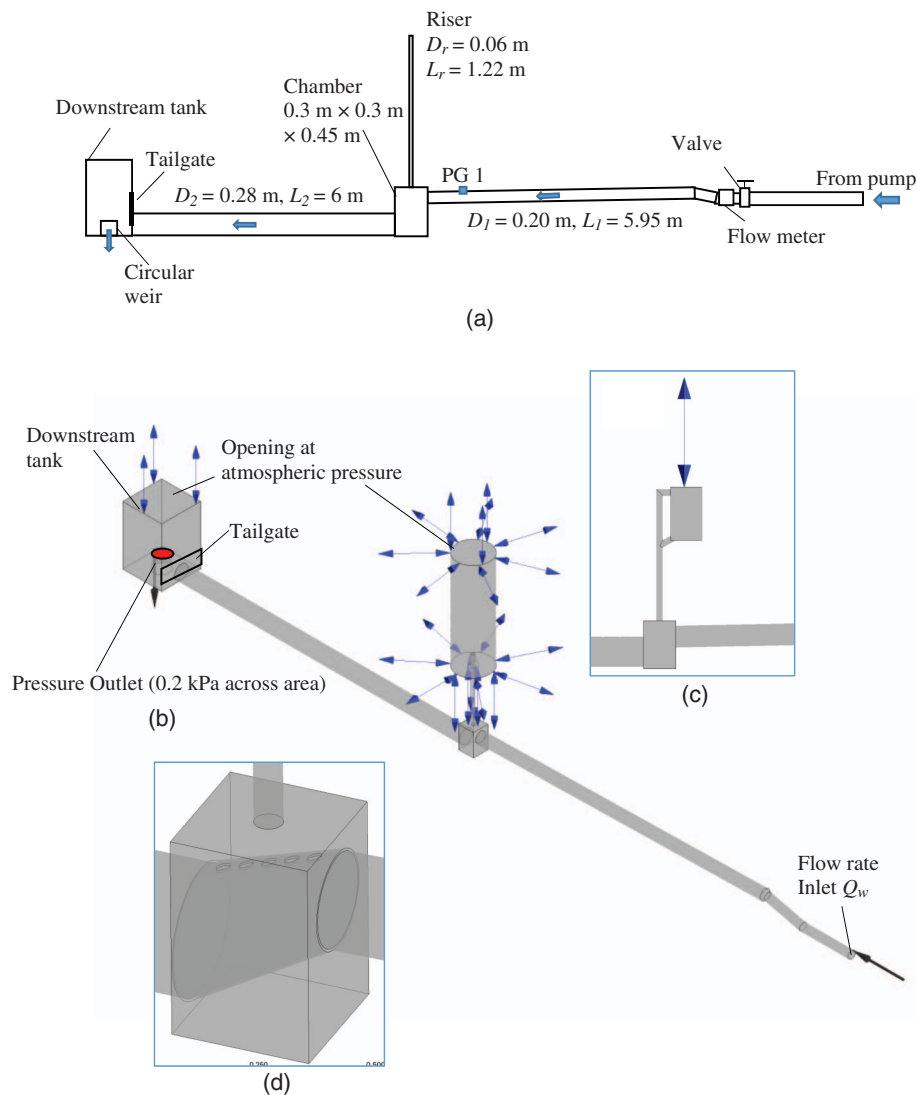


Fig. 2. (Color) Geometry of the numerical model: (a) schematic of the physical model (drawing not to scale); (b) schematic of the geometry and boundary conditions of the numerical model; (c) model with water recirculation chamber connected to the riser; and (d) model with benching (with five openings on its top wall) in the chamber.

The valve utilized in the experiment was a butterfly valve that could be fully opened from fully closed by a quarter turn. The valve was operated manually, and valve opening time was kept as consistent as possible. In the numerical model, the water inflow rate was defined as increasing from the initial to the final flow rate linearly in 0.1 s.

Pressure was measured at the upstream pipe crown and 0.3 m upstream from the chamber (PG1) using a pressure transducer (OMEGA MMCG015BIV-10P4C0T3A5CE; OMEGA Engineering, INC.; Norwalk, Connecticut) with an accuracy of $\pm 0.2\%$ and a range of ± 130 kPa. The water inflow rate was measured by a Foxboro (IMT31A, Schneider Electric Systems, Foxboro, Massachusetts) magnetic flow meter. The sampling frequency for the pressure transducer and the flow meter was 1,000 Hz. A more detailed description of the experiments and the related physical processes can be found in Liu et al. (2020).

Fig. 2(b) shows the geometry of the numerical model; the size of the numerical model was the same as the size of the physical model. In the numerical model, the open space above the riser was modeled as a cylindrical block in order to better define boundary conditions as atmospheric pressure. The boundaries of the cylindrical

block and the top boundary of the downstream tank were defined as atmospheric pressure. The outlet boundary of the system was defined as a pressure outlet at the top opening of the circular weir in the downstream tank, as illustrated in Fig. 2(b). The pressure at the outlet boundary was set at 0.2 kPa across the outlet area in order to match the simulated water level in the downstream pipe and in the chamber with the measured level in the physical model. The pressure specified at the outlet boundary had no significant effect on the geyser procedure in this study, according to a sensitive analysis. In the current air/water two-phase model, air was defined as an ideal gas that follows the ideal gas law, and water was considered to be incompressible.

The potential retrofitting methods of water recirculation chambers and benching listed in Table 1 are shown in Figs. 2(c and d). The WRC shown in Fig. 2(c) was located beside the riser at the top. The chamber had a diameter of 0.3 m and a height of 0.45 m. It was connected to the original riser with two pipes with the same diameter as the riser by a 90° bend at the riser top and a 30° sloped pipe at the chamber bottom. The top of the chamber was defined as the atmospheric pressure. Fig. 2(d) shows the benching installed in the chamber. The benching was a variable-diameter pipe segment

Table 1. List of experiments

Run series	Initial inlet, Q_w (L/s)	Final inlet, Q_w (L/s)	Outlet BC (kPa)	Initial water level in riser ^a	Tailgate opening	Mitigation method
A1 ^b	20	85	0.2	No water in riser	Fully open	No mitigation
A2 ^b		40		0.41 m	0.055 m	No mitigation
B1	20	85	0.2	No water in riser	Fully open	WRC
B2						Benching with openings
B3						OP: $d_0 = 0.5D_r$
C0	20	40	0.2	0.41 m	0.055 m	OP: $d_0 = 0D_r$
C1						OP: $d_0 = 0.1D_r$
C2						OP: $d_0 = 0.2D_r$
C3						OP: $d_0 = 0.4D_r$
C4						OP: $d_0 = 0.8D_r$
D1 ^c	20	100	N/A	No water in riser	Fully open	$T = 0.2$ s
D2 ^c						$T = 10$ s
D3 ^c						$T = 20$ s
E1 ^c	20	40		0.30 m	0.055 m	$T = 0.2$ s
E2 ^c						$T = 20$ s
E3 ^c						$T = 35$ s

Note: WRC = water recirculation chamber; OP = orifice plate; d_0 = diameter of the center opening; and T = upstream valve opening time.

^aMeasured from the bottom of riser. All experiments in the A, B, and C runs had $T = 0.1$ s; values for the D and E runs are given in the table.

^bModel validation.

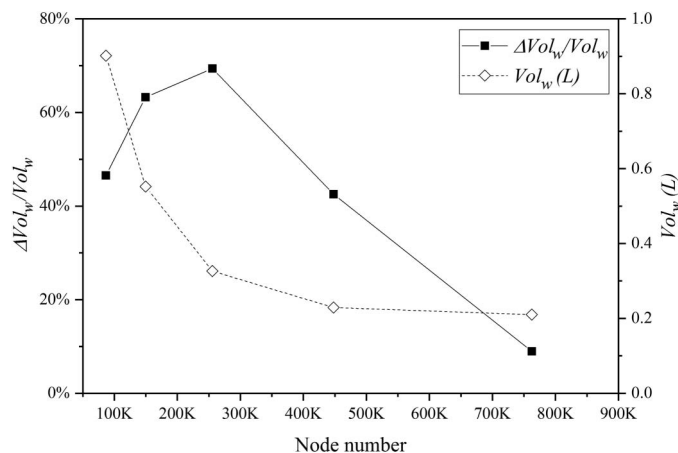
^cPhysical experiments.

that connected the upstream and downstream pipes. Five openings with diameters of 0.03 m each were cut at the top of the benching for ventilation. The inflow control method was carried out by slowly opening the upstream valve, representing a controlled increase in the water inflow rate.

A commercial software, Ansys CFX version 16.2, was used to simulate the process of geyser events and the effectiveness of potential solutions for mitigating the geyser events. The software is known for its stability and efficiency in solving 3D CFD problems. Results from previous physical model studies were used for the calibration and validation of the numerical model. In the simulation, the Reynolds-averaged Navier-Stokes (RANS) model was used to simulate the transient flow process. The standard $k-\varepsilon$ turbulence model was used to model the turbulent flow. Two equations were added to the governing equations for solving the turbulent kinetic energy and dissipation rate. The $k-\varepsilon$ model is able to reflect turbulent features without adding much computing time compared to the large eddy simulation (LES) or direct numerical simulation (DNS) models. For multiphase flow modeling, the volume of fluid (VOF) method (Hirt and Nichols 1981) was used. This method introduces an additional variable α to represent the volume fraction of each phase in each control volume.

The mesh size for the geometry was determined by the mesh independence test. The mesh was locally refined in the riser and chamber. A total of 760,000 (760K) nodes were generated. A mesh independence test was conducted, and the results are shown in Fig. 3. The total volume of water flowing out of the riser (Vol_w) was selected as the indicator of the test, and the simulation was conducted using a series of progressively refined mesh levels. The change in the total volume of water flowing out of the riser was within 10% at 760K nodes. Further increasing the node number did not improve the results significantly. Therefore, the 760K mesh level was selected for the simulations. All simulations were then conducted at this mesh level.

The time step was initially set at 0.005 s in order to maintain the Courant number at around 1.0 at this mesh level. Based on the sensitivity analysis for the time step, a shorter time step of 0.001 s was used during geyser events. During CFD modeling, we also used a time step of 0.001 s at the beginning of the simulations. The peak

**Fig. 3.** Mesh independence analysis.

pressure was found to be at about 2.0 kPa, which was only slightly lower than the experimental result (approximately 2.5 kPa). It seems that the initial pressure change was not sensitive to the time step. The simulation was conducted using a Dell workstation (Dell, Round Rock, Texas) with an Intel Xeon E5-2450 (Intel, Santa Clara, California) processor. Four cores were used during one simulation, and the computing time at this mesh level for each case was about one week. When using 10 cores, the execution time was reduced to 4 days for each run.

In the prototype system, the upstream pipe had a smaller diameter than the downstream pipe. However, the downstream pipe had a complicated connection in which it bifurcated into two smaller pipes before discharging to the river. The city's MikeUrban model also suggested that the downstream pipe gets surcharged before the upstream pipe. The simulation cases selected were associated with Liu et al. (2020). In the laboratory study, a variety of combinations of upstream and downstream flow conditions was tested. No geyser was observed for the downstream open channel flow case, and this flow scenario was, therefore, omitted from this geyser mitigation study.

Table 1 summarizes all of the simulated cases. The experimental results were applied to validate the CFD models for Runs A1 and A2.

1. Run A1 was a geyser triggered by a surge front while the downstream pipe was near full, and Run A2 was a geyser triggered by the release of an air pocket. The downstream water levels were controlled by the circular overflow weir and the sluice gate, respectively, for Runs A1 and A2.
2. Runs B1–B3 tested different mitigation methods for geysers triggered by surge fronts while the downstream pipe was near full (Run A1). For Run B3, an orifice plate was installed in the riser, 0.76 m above the bottom of the riser.
3. Runs C tested the effectiveness of an orifice plate in mitigating geyser events triggered by the release of an air pocket (Run A2). For Runs C1–C4, the orifice plate was located at the top of the riser. Run C0 was for testing the riser fully sealed at the top, based on the boundary condition of B2.
4. Runs D and E were conducted in the physical model to test the effect of the upstream valve opening time in mitigating geysers in which the increase in the water flow rate was controlled by the upstream valve opening time.

For Runs A1, B1–B3, and D1–D3, the tailgate was fully opened. For Runs A2, C0–C4, and E1–E3, the tailgate was partially closed, which restrained the flow capacity in the downstream pipe. The initial water level in the riser was 0.41 m from the bottom of riser. The initial inflow rate was 20 L/s, and an air pocket with a volume of 0.071 m³ was predefined in the upstream pipe, 0.5 m upstream of the chamber. In the physical model, the tailgate was initially fully open, with stable open channel flow in the pipes and chamber. The tailgate was then lowered, and a hydraulic jump propagated upstream and trapped an air pocket in the upstream pipe.

Results and Discussions

Model Validation

Run A1 studied a geyser triggered by a surge front while the downstream pipe was near full. The flow in the downstream pipe was almost a full pipe flow, with a thin layer of air along the pipe crown. The initial air flow rate in the downstream pipe was 0.9 L/s in the downstream direction, and the initial water inflow rate was 20 L/s. The flow in the upstream pipe was an open channel flow with a water depth of 0.08 m. A sudden increase in the water inflow rate caused a rapidly filling flow with a surge front in the upstream pipe and generated geyser events; this was observed both in the physical experiment and in the numerical simulation. A steady-state simulation with a water inflow rate (Q_w) of 20 L/s was conducted prior to the transient simulation. For the initial steady-state simulation, the measured water depth in the chamber of the physical model was 0.24 m, and the simulated pressure at the bottom of the chamber was 2.66 kPa, which indicated a water depth in the chamber of around 0.27 m. The numerically simulated pressure showed agreement with the measured data.

Fig. 4(a) shows the simulated and measured pressures at PG1 for Run A1. The numerical simulation demonstrated a pressure process similar to that of the physical measurement and agreed well with the physical measurements. Specifically, the pressure started to increase sharply at $t = 2.0$ s, which was the moment at which the water front reached the chamber and completely choked the downstream pipe, as shown in Fig. 4(b). The simulated pressure reached its maximum of 7.89 kPa at $t = 2.43$ s, which was about 6.3% lower than the measured 8.43 kPa at $t = 2.37$ s. From $t = 2.50$ s to 2.70 s, the geyser event began when the air/water mixture

was ejected out of the riser and the simulated pressure decreased due to the release of the air/water mixture. After the geyser event, the simulated pressure in the upstream pipe started to increase again because of the continuous incoming flow from upstream. The peak pressure was 2.9 kPa, about 20% different from the measured peak of 3.6 kPa.

The major discrepancy between the experimental and numerical simulations was that there was a pressure increase at the beginning of the simulations, within 0.50 s. This was mainly due to the instability of the numerical model when dealing with sudden changes in the water inflow rate at the beginning of the simulation. However, the numerical and experimental processes after the onset of the geysers still agreed with one another, which indicates that the analysis of the geysers was valid. Therefore, the numerical model could be used to simulate geysers triggered by surge fronts while the downstream pipe was almost full.

Fig. 4(b) shows the simulated volume fraction of water for Run A1. It can be seen that at $t = 0$ s, the air in the upstream pipe and in the chamber was connected with the atmosphere via the riser. The surge front in the upstream pipe propagated at a speed of about 2.6 m/s, which was close to the physically measured rate of 2.5 m/s. When the wave front arrived at the chamber, it impacted the opposite chamber wall and curved back ($t = 2.20$ s). With the continuous incoming flow, the water level in the chamber increased, and the pressure at PG1 also began to increase at $t = 2.00$ s. In Fig. 4(a), the pressure reached a maximum at $t = 2.40$ s. After the air/water mixture erupted out of the riser, the pressure dropped to a minimum at around $t = 2.65$ s. The water eruption in Run A1 was similar to the inertial oscillation described in Guo and Song (1990); however, in this case the flow was not a single-phase water flow but was an air/water mix with a lower average density.

Geyser height is mainly related to the peak pressure inside the chamber and the upward momentum of the air/water mixture induced by the pressure. For a sudden increase in the inflow rate from 20 to 85 L/s, Fig. 4(a) shows that the peak pressure in the upstream pipe reached 8.0 kPa. Given that the height of the riser was 1.22 and 0.05 m from the pipe crown to the chamber top, the pressure was not enough to push a column of pure water out of the riser. The observed geyser height in the physical model was 0.15 m above the riser top, which compares well with the simulated height of 0.20 m. This was in agreement with the field results reported in Wright et al. (2011), which indicated that the pressure head in a tunnel does not have to be as high as ground level in order for a geyser to occur. This is likely due to the fact that if the density of the mixture is less than pure water, it can, therefore, be driven to a higher elevation. For Run A1, the amount of water ejected out of the riser was 0.21 L, based on the simulation, and the amount of air transported downstream was 13.6 L.

For Run A2, the geyser was triggered by an entrapped air pocket with water in the riser. For this type of geyser, both the physical experiment and the numerical simulation were conducted with an initial water inflow rate $Q_w = 20$ L/s and a final inflow rate $Q_w = 40$ L/s. In the initial condition, the air pocket in the upstream pipe was 0.5 m upstream from the chamber, and the length and total volume of the air pocket were 0.7 m and 0.071 m³, respectively. The initial pressure inside the air pocket was 2.7 kPa, based on the physical measurements. The initial water level in the riser was 0.41 m from the bottom of the riser. The measured and simulated pressures at PG1 are shown in Fig. 5(a). In the experiment, the measured pressure at PG1 increased rapidly from the initial value of 4.4 to 13.3 kPa and caused the water column to rise in the riser and erupt out. The pressure then decreased to a minimum of 1.7 kPa at 2.00 s. The pressure then increased back to about 11.6 kPa.

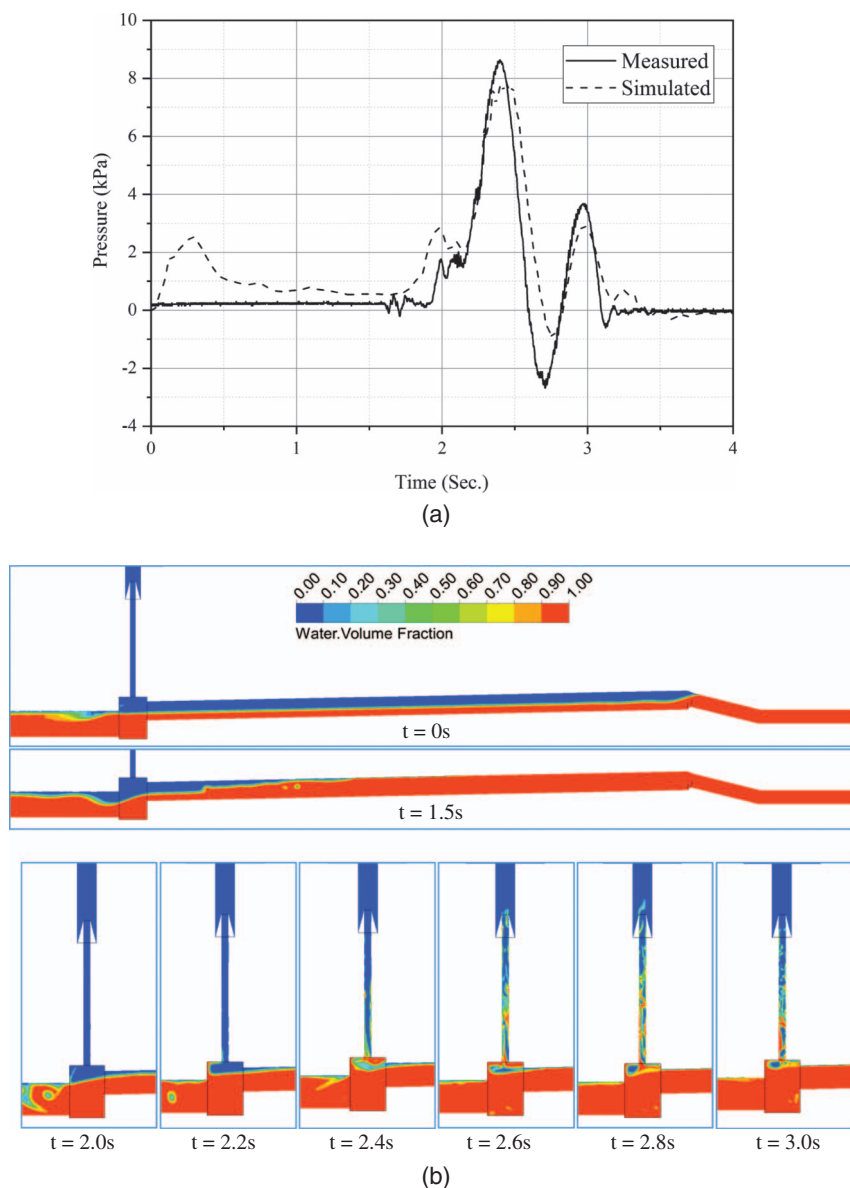


Fig. 4. (Color) Simulated geyser event triggered by a surge front while the downstream pipe is almost full: (a) simulated and measured pressure at PG1; and (b) simulated volume fraction of water.

Fig. 5(a) also shows that the simulated pressure at PG1 agreed well with the measured pressure, especially before $t = 3.00$ s. The simulated maximum pressure was 13.8 kPa, which was slightly higher than the measured pressure of 13.3 kPa. The minimum simulated pressure was 0.9 kPa, and the minimum measured pressure was 1.7 kPa. The simulated maximum and minimum pressures were within 20% of the measured ones. The onset of the maximum and minimum pressures for the numerical and physical models were in good agreement with each other. This study focuses only on the most severe geyser, which happened before $t = 5.00$ s. Fig. 5(b) shows the water volume fraction for Run A2. It can be seen that the water column in the riser was pushed out of the riser by the air pocket from the upstream pipe, beginning after $t = 0.70$ s and finishing at $t = 1.10$ s. After the water column was expelled from the riser, PG1 was connected to the atmospheric pressure, and the pressure decreased to almost 0 Pa. Then the water front in the upstream pipe reached the chamber again and caused the blockage of air flow, and the pressure increased again after $t = 1.5$ s. Between $t = 1.10$ s and 3.00 s,

no water erupted out of the riser due to the lower pressure in the chamber after the onset of the first geyser.

For Run A2, the simulation results suggested that the amount of water ejected out of the riser before $t = 3.00$ s was 3.5 L, which was much higher than the amount ejected in Run A1. The air transported downstream was 0.32 L, which was lower than the amount transported in Run A1. This was mainly because, in Run A2, the water in the riser blocked the air from being transported to the downstream pipe.

Potential Ways to Mitigate Geysers Triggered by Surge Fronts while the Downstream Pipe Is Near Full

Run B studied the effectiveness of three geyser mitigation methods: WRCs, benching, and orifice plates. A geyser triggered by a surge front while the downstream pipe was near full was selected for testing these mitigation methods because of its distinctive pressure pattern and relatively short computing time. Fig. 6 shows the volume fractions of water when the different mitigation methods were

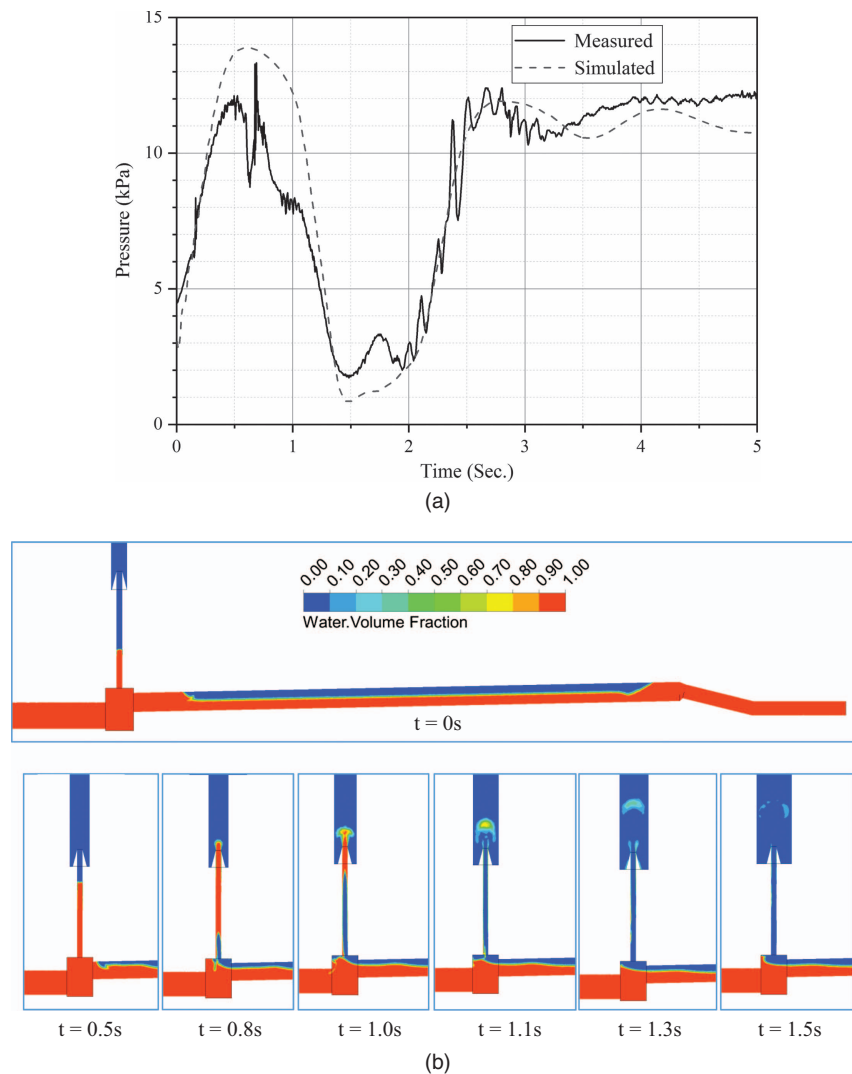


Fig. 5. (Color) Simulated geyser event triggered by an entrapped air pocket with water in the riser: (a) simulated and measured pressure at PG1; and (b) simulated volume fraction of water.

applied to the geyser from Run A1. Fig. 6(a) shows the geyser event when the WRC was installed (Run B1). The occurrence of geyser event was similar to the one in Run A1. The surge front arrived at the chamber at $t = 2.0$ s and curved backward, which blocked the bottom of the riser. Once the air/water mixture reached the riser top, the air/water mixture was redirected by the 90° bend toward the WRC instead of directly shooting upward at $t = 2.5$ s. Therefore, no visible geyser was noticed. The air and water were automatically separated in the WRC chamber. Air escaped from the WRC chamber via its top opening and water flowed back to the sewer system via riser, as shown in Fig. 6(a), at $t = 3.00$ and 4.00 s, respectively. The chamber of the WRC temporarily stores the air/water mixture during a geyser event. Therefore, when designing a WRC for prototype storm sewer systems, the volume of the WRC chamber should be sufficient to collect the water. The connection pipes can be designed to be the same diameter as the manhole shaft for an effective redirection of the air/water mixture and for receiving the water that flows back to the riser. Other design parameters, such as the footprint area, depth, distance to the riser, and so forth, were not tested in this study and should be carried examined in future studies.

Fig. 6(b) shows the geyser process when benching was installed (Run B2). No visible geyser occurred. More specifically, at the

beginning of the simulation, the upstream pipe had an open channel flow. Air/water mixture was observed at the connection of the chamber and the downstream pipe. When the front advanced toward the chamber, the air pocket that initially occupied the upper portion of the upstream pipe was compressed due to the small openings on the benching. The water surface in the downstream pipe and benching was lowered from $t = 0.50$ to 2.00 s by this higher pressure caused by the compressed air pocket. At $t = 2.00$ s, the water front reached the chamber. After $t = 2.50$ s, water inside the benching shot out through the opening holes on the benching top and only a portion of the air/water mixture flowed into the riser. The momentum of the air/water mixture was not great enough to drive the mixture out of the riser. No geyser was observed in this case.

Fig. 6(c) shows the geyser event when an orifice plate with an opening $d_0 = 0.5D_r$ (where D_r = riser diameter) was installed inside the riser, 0.76 m above the chamber top. As in the benching case, the water level in the chamber decreased at the beginning, and the downstream pipe became choked at around $t = 2.5$ s. The upward motion of the air/water mixture was blocked by the orifice plate, and therefore, the geyser was mitigated by the orifice plate.

Fig. 7 shows the simulated pressure variations at PG1 for the foregoing cases compared with the values for Run A1. The pressure

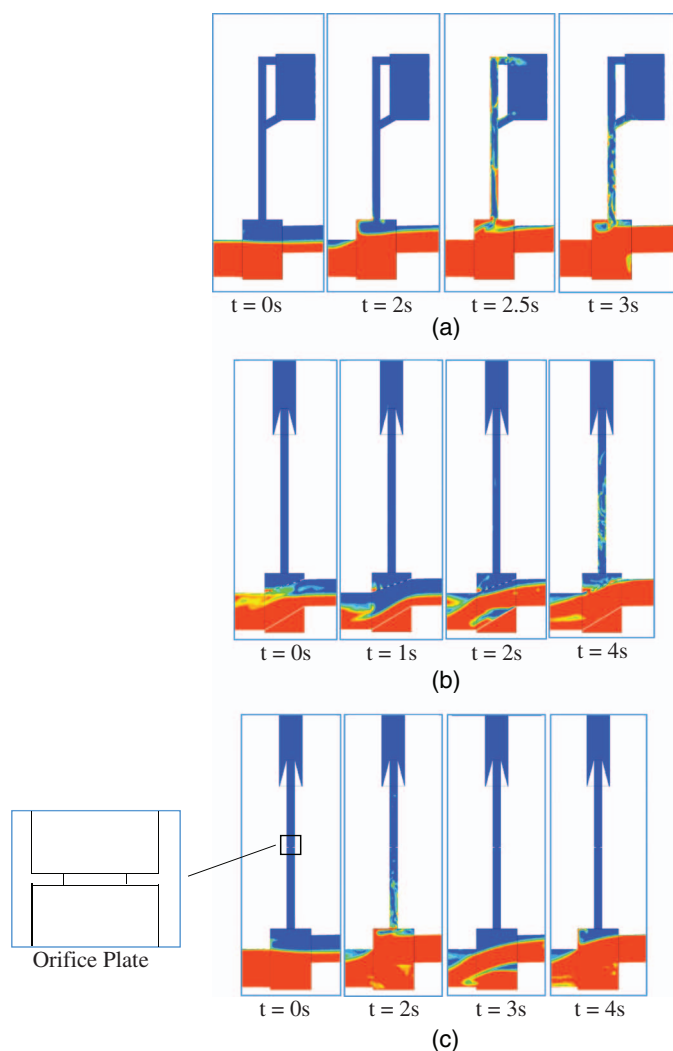


Fig. 6. (Color) Simulated volume fraction of water for different mitigation methods for a geyser triggered by a surge front while the downstream pipe is almost full: (a) water recirculation chamber; (b) benching with openings; and (c) orifice plate ($d_0:D_r = 0.5$).

pattern at PG1 for the case with the WRC (Run B1) was similar to the pattern for Run A1. The peak pressure for Run B1 was 10 kPa, slightly higher than the one without a WRC. The increased peak pressure was mainly due to the resistance caused by the additional elbow and related structures at the top of the riser. With regard to the simulated water volume fraction, no water was observed flowing out of the WRC chamber [Fig. 6(a)]. This was mainly due to the redirection of the air/water mixture at the top of the riser.

For Runs B2 and B3, the installation of the benching and the orifice plate increased the pressure at PG1 at the early stage of the simulation ($t < 1$ s). This was due to the blockage of airways caused by the benching and the orifice plate. As the surge front propagated downstream to the chamber, the orifice plate or benching acted as obstructions and increased the head loss of air flow in the riser. Therefore, a higher pressure was needed for the propagation of the surge wave front to push the air out of the openings. The increased pressure was high enough to lower the water level in the benching and chamber, as shown in Figs. 6(b and c), respectively. The decreased water level in the benching and chamber connected the air pocket in the upstream and downstream pipes. Air is likely to be transported downstream and can cause geyser events in

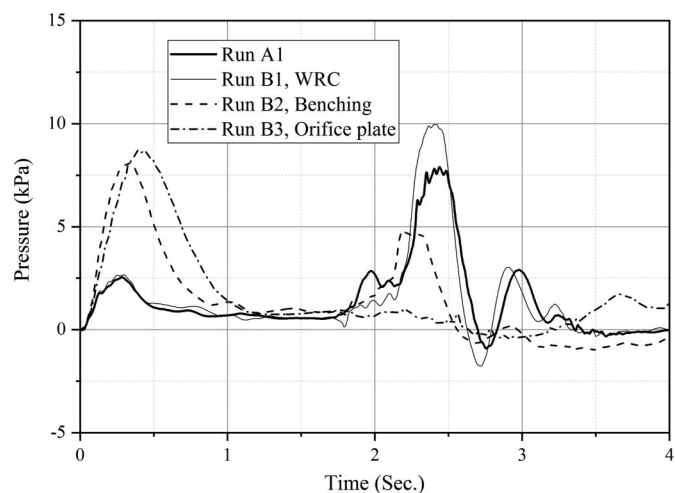


Fig. 7. Simulated pressure at PG1 for geyser mitigating methods with a geyser triggered by a surge front while the downstream pipe is almost full.

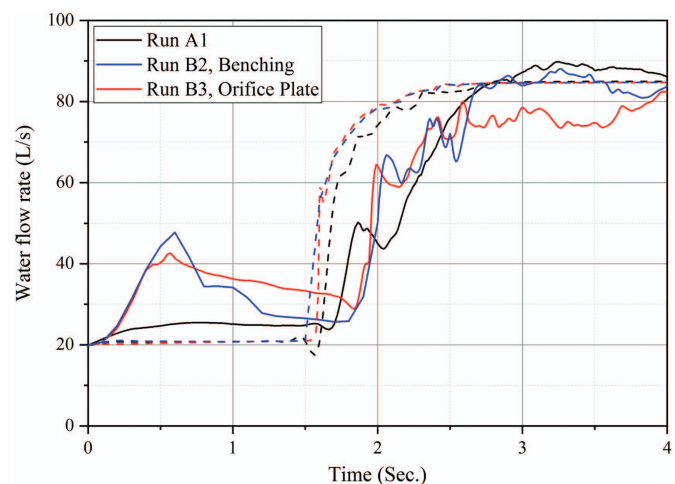


Fig. 8. (Color) Hydrograph of water inflow and outflow from the chamber. Solid line = flow rate in the downstream pipe; and dashed line = flow rate in the upstream pipe.

downstream areas. During the geyser event, from $t = 2.00$ to 3.50 s, the benching with openings on the top and the orifice plate significantly reduced the peak pressure from the original 8.0 kPa to slightly less than 5.0 and 0.8 kPa, respectively. The benching and orifice alternatives also offered more energy loss for the water flow upward toward the vertical shaft. No visible geyser was observed during the simulations in which the benching and orifice plate were installed (Runs B2 and B3).

Fig. 8 shows a hydrograph of the water flow rate into and out of the chamber during the geyser event with mitigation using benching and an orifice plate (Runs B2 and B3). The flow in the upstream pipe increased from 20 to 85 L/s. The flow in the downstream pipe, due to the benching or orifice plate, increased at rate similar to the increase in Run A1. The benching guided the flow more smoothly to the downstream tunnel and, therefore, decreased the turbulence in the chamber, causing a different pressure variation during the geyser event.

It would be prudent to minimize the amount of air transported downstream in prototype sewer structures in order to avoid potential geyser events in downstream areas. In the current numerical

Table 2. Summary of different geyser mitigation methods based on the CFD model

Run	Mitigation method	Maximum pressure at PG1 (kPa)	Amount of water flowing out of the riser (L)	Amount of air transported downstream (L)
A1	No mitigation	8.0	0.21	13.6
B1	WRC	10.0	No geyser	14.1
B2	Benching	5.0	No geyser	38.3
B3	OP: $d_0 = 0.5D_r$	0.8	No geyser	55.0

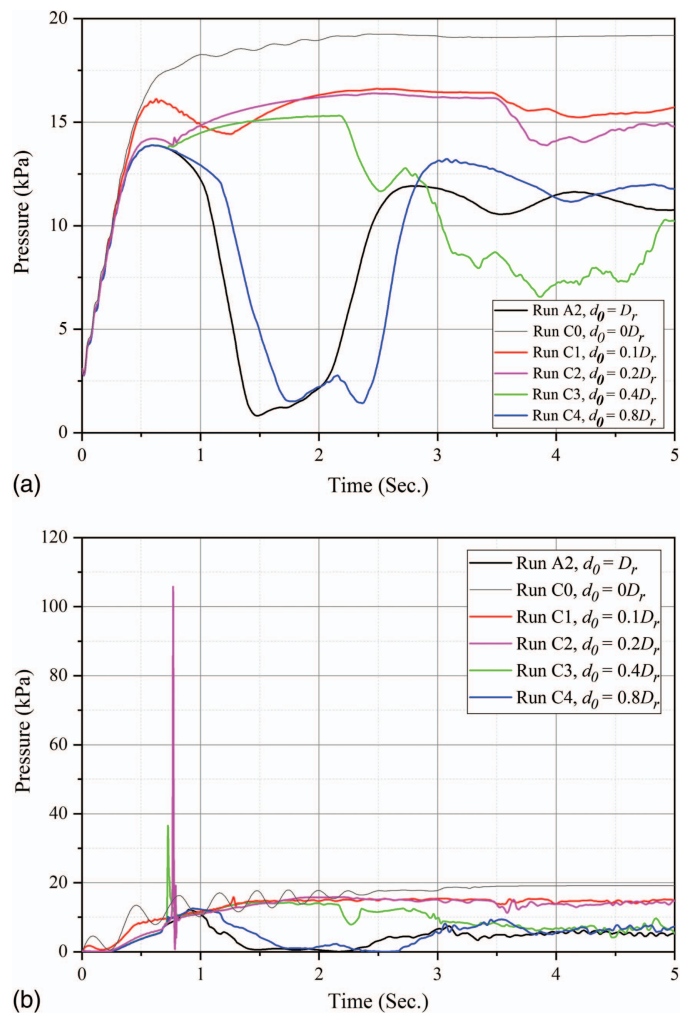
model, the air flow rate at 0.35 m downstream of the chamber was integrated with time in order to estimate the total amount of air transported downstream. For Run A1, the total simulated amount of air transported downstream was 13.6 L. In Runs B1, B2, and B3, a total of 14.1, 38.3, and 55.0 L of air were transported downstream with the WRC, benching, and orifice plate, respectively. Compared to the original geyser event without mitigation, benching and orifice plates can potentially result in an increased amount of air being transported downstream. By adding benching or an orifice plate, the airway through the riser was restricted and, therefore, more air tended to be transported downstream.

Table 2 summarizes the performance of the different geyser mitigation methods on the geyser triggered by a surge front while the downstream pipe was almost full. The orifice plate was able to efficiently reduce the peak pressure and the amount of water flowing out of the riser during the geyser event. However, the air transported downstream increased by 4 times with the orifice plate installed; this could cause geyser events at downstream locations. The initial air volume in the system was estimated to be 55 L, including the air pockets in the upstream pipe, the chamber, and the riser. At the location of the geyser in Edmonton, a new tunnel is connected to an existing storm sewer. The additional flow from the new tunnel may cause issues when the existing pipe is already full during design rainfall events. It is obvious that, with the orifice plate method, almost all air previously in the system is transported downstream, which could trigger geyser events further downstream. The other mitigation alternatives transported a fraction of the air previously in the system downstream.

Orifice Plate for Mitigating a Geyser Triggered by an Entrapped Air Pocket

As discussed previously, an orifice plate can decrease pressure in the upstream pipe and mitigate a geyser event. The effectiveness of orifice plates was further examined in the C runs with a geyser triggered by an entrapped air pocket. A riser fully sealed case and orifices with four different openings were tested (with $d_0 = 0.1D_r$, $0.2D_r$, $0.4D_r$, and $0.8D_r$). Figs. 9(a and b) show the simulated pressures at PG1 and in the middle of the riser at 0.76 m above the bottom of the riser. The pressure at PG1 increased slightly for Runs C0, C1, C2, and C3, with maximum pressures at 19.2, 16.6, 16.3, and 15.3 kPa, respectively. Compared with the pressure of 13.9 kPa for Run A2, the pressure at PG1 generally increased with the installation of an orifice plate. Smaller orifices corresponded with higher pressure, and the riser fully sealed case generated the highest pressure at PG1. The pressure variation for Run C4 was similar to the variation for Run A2. This was because the orifice size for Run C4 was $0.8D_r$, which was close to the diameter of the riser, and the effect of the orifice plate was minor in comparison with the other orifice sizes.

Fig. 9(b) shows the pressure in the middle of the riser at 0.76 m above the bottom of the riser. For Run C0, an oscillating pressure

**Fig. 9.** (Color) Simulated pressure variations for Runs C1–C4: (a) simulated pressure at PG1; and (b) simulated pressure on the orifice plate installed inside the riser.

was observed. This was mainly due to the compression and expansion of the air pocket in the riser above the water column. Fig. 9(b) also shows a sharp peak for Runs C1–C3. This was primarily because of a sudden slamming of the water slug on the orifice plate. The figure suggests that the simulated peak pressures in the riser for Runs C1, C2, and C3 were 15.9, 105.8, and 36.6 kPa, respectively, and the pressure at PG1 after the transient flow happened was about 14 kPa. The peak pressure was mainly induced by a sudden stop of the water column when it reached the orifice plate. The trend for pressure agreed well with Zhou et al. (2002), who found that maximum peak pressure occurred at $d_0 = 0.2D_r$. However, the pressure was less than that the pressure reported by Li and Zhu (2018), mainly because the discharge through the orifice was a lower-density air/water mixture rather than a single-phase water discharge. Small air bubbles in the mix can reduce acoustic celerity (Wylie and Streeter 1993). An air pocket under a water column can add overall elasticity to the system but does not necessarily change the acoustic wave speed within the water column itself (unless it consists of an air/water mix). The increased elasticity in the system provides a cushion for the compression and expansion of air and, therefore, decreases the pressure. Normalized by the peak pressure of Run C1, the ratios of the peak pressure were 1.00, 6.65, and 2.30, respectively, for Runs C1, C2, and C3. The absence of spikes at PG1 was mainly because it

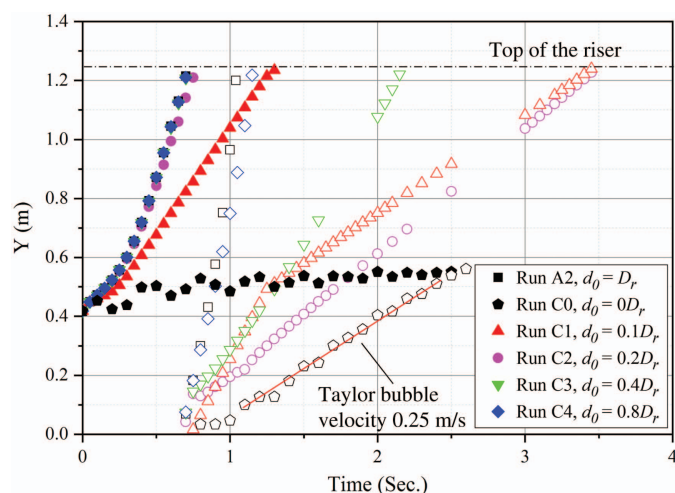


Fig. 10. (Color) Simulated top (solid symbols) and bottom (void symbols) of the water column.

was located at the pipe side wall, where pressure was only affected by the static pressure of fluid in the pipe, while the spikes that occurred locally at the orifice plate for C2 and C3 were mainly due to the impingement of the water column on the orifice plate. It may also be due to the fact that the air trapped between the water column and the chamber damped the pressure variations at the pressure transducer in Run C1.

Fig. 10 shows the simulated movement of the top and bottom of the water column in the riser during the geyser event. The origin of the coordinate Y is set at the bottom of the riser. The figure shows that, except for Runs C0 and C1, the simulated top of the water column overlapped with each other and reached the top of the riser at around 0.7 s, at which point a sharp peak in pressure was observed, as shown in Fig. 9(b). The velocity of the water column was about 2 m/s. For Run C0, the top of the water column fluctuated well with the pressure shown in Fig. 9(b). It indicates that the elevation variation of the top of the water column compressed and expanded the air pocket and, therefore, generated an oscillating pressure at the sealed plate. The bottom of the water column for Run C0 rose, and the velocity of the movement corresponded well with the Taylor bubble velocity, for which the velocity can be written as $0.345\sqrt{gD_r} = 0.25$ m/s. For Run C1, because of the limited capacity for air to flow out of the orifice, the top of the water column moved upward much more slowly than in the other cases and reached the orifice plate at around 1.20 s, which corresponds with the small peak pressure in Fig. 9(b). Run A2 corresponded to the fastest-moving bottom of the water column; the interface reached the top of the riser at 1.04 s. Run C4 had minor differences compared with Run A2, mainly due to the limited effect of the orifice plate of $0.8D_r$. The bottom of the water column reached the top of the riser at around 3.45, 3.45, and 2.00 s, respectively, for Runs C1, C2, and C3.

Currently, there is no hydrograph on the inflow rate, and the existence of air pockets in the prototype system is hypothetical. The following analysis provides a rough estimation of the movement of the water column and geyser height. Following the momentum equation and variations in the simulated pressure at PG1, the water level movement in the riser for Run A2 can be calculated using

$$dpA = \frac{d(mv)}{dt} \quad (1)$$

where A = cross-sectional area of the riser; p = pressure at PG1, assuming the pressure in the chamber equals the pressure at PG1;

m = mass of the water column in the riser; v = velocity of the water column in the riser; and t = time.

It is assumed that the water column in the riser initially is pushed up by the pressure at PG1 with negligible friction and minor losses. Based on the measured pressure variations at PG1 [as in Fig. 5(a)], the results obtained from Eq. (1) suggest that the top of the water column reaches the riser top at 0.51 s, which is close to the simulated value of 0.70 s. The velocity of the water column is 3.52 m/s when its top of the water column the top of the riser. This is also close to the simulated value of 2 m/s. The geyser height corresponding to this velocity of 3.52 m/s is 0.63 m, which is in agreement with the simulated value of 0.60 m. This indicates that this analysis could be applied to prototype systems once the pressure variations in the prototype are known. Scaled to the prototype with a scale of 20:1, the geyser height in the prototype system is about 12 m.

Calculated from Fig. 10, the average velocity of the top of the water column reached the orifice plate was 0.4, 1.38, 1.4, and 1.68 m/s for Runs C1–C4, respectively. The average velocity of the water flow under the orifice plate after the impingement was 0.12, 0.18, 0.84, and 1.75 m/s for Runs C1–C4. Given lengths of the water column of 0.63, 1.07, 1.13, and 1.14 m for Runs C1–C4 from Fig. 10, the momentum changes for Runs C1–C4 were 0.42, 3.06, 1.79, and -0.22 kg · m/s, respectively. The negative momentum change was mainly because the effect of the orifice plate with $d_0 = 0.8D_r$ was small, and the water column was still accelerating due to the compressed air pocket. Normalized by the momentum change in Run C1, the ratio of the momentum changes for Runs C1, C2, and C3 were 1.00, 7.28, and 4.26, respectively, which is generally in agreement with the ratios of pressure discussed previously. Therefore, it can be concluded that the peak pressure was mainly induced by the impingement of the water column on the orifice plate.

Fig. 11 shows the volume fraction of water in the vicinity of the orifice plate during the geyser event. Geyser height is defined as the maximum displacement of the water column above the riser top, and the change in the morphology of the water from a continuous flow to dispersed droplets cannot be achieved by the current numerical model. Therefore, the maximum displacement of the water column was used as a measure of the height of the geyser. In the numerical model, it was found that, for Runs C1–C3, the water shot out of the orifice plate continuously and maintained a geyser height similar to a circular jet. Therefore, the geyser height for Runs C1, C2, and C3 was approximated as the height of the jet. Compared to the geyser height of 0.6 m for Run A2, the height of the geyser was significantly reduced by the orifice plate. Fig. 11 shows the maximum geyser height for Runs C1, C2, and C3 at 0.08, 0.11, and 0.18 m, respectively; the geyser reached 0.60 m for Run C4, because the effect of the orifice plate was not significant.

The geyser triggered by an entrapped air pocket with water in the riser tended to have a relatively small amount of air transported downstream compared with the geyser triggered by a surge front while the downstream pipe was almost full. The numerical model suggested that in Run A2, the total amount of air transported downstream was 0.32 L, which is an order of magnitude smaller than the amount in Run A1. For Runs C0, C1, C2, C3, and C4, the total amount of air transported downstream before $t = 3.00$ s was 1.83, 1.63, 1.46, 1.13, and 0.36 L, respectively. This was in agreement with the previous discussion; more restriction of the air flow at the riser results in a higher amount of air transported downstream.

Table 3 is a summary of the performance of the orifice plates studied in this section. The installation of an orifice plate can significantly reduce geyser height and the amount of water flowing out of the riser. Sealing the riser can prevent geyser events completely

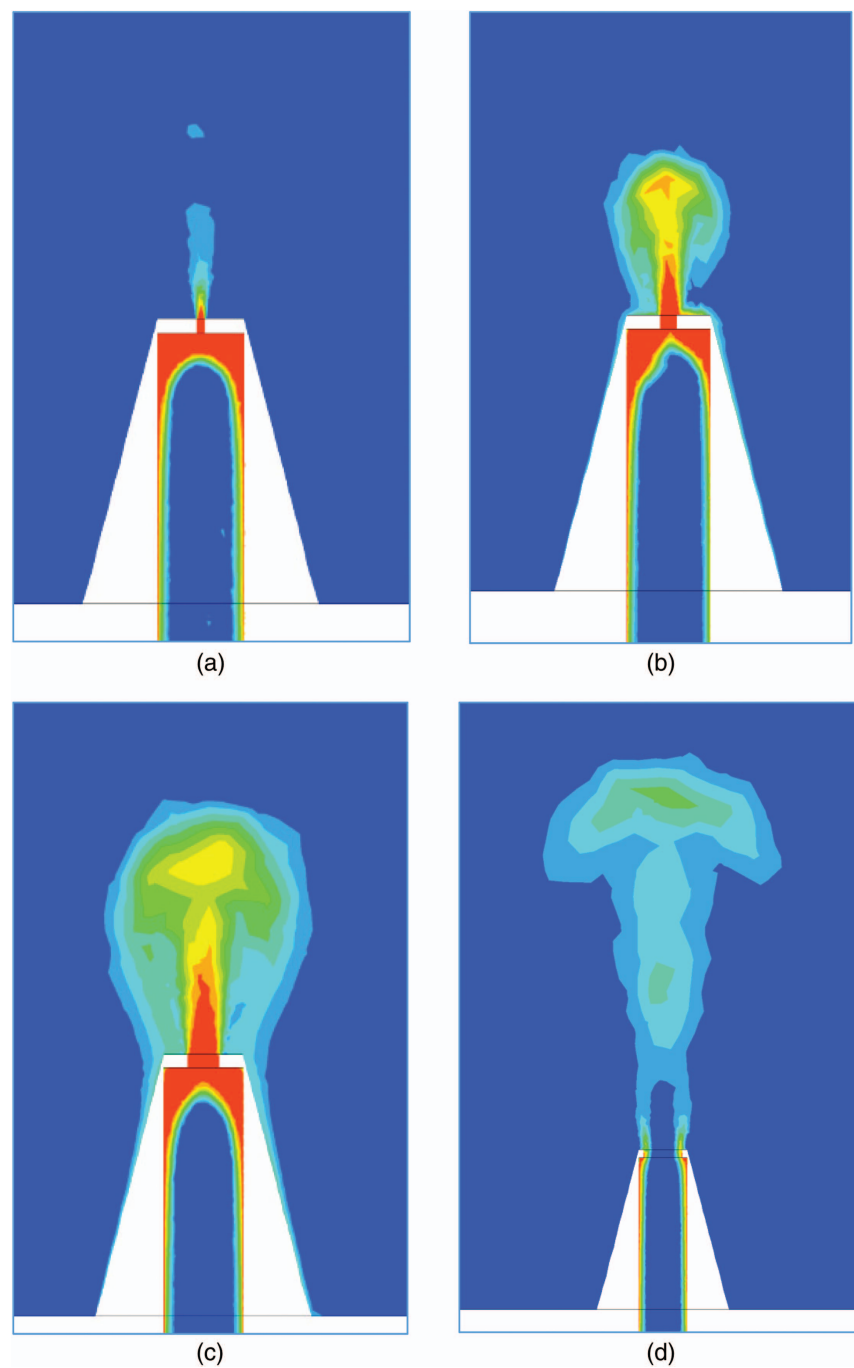


Fig. 11. (Color) Simulated volume fractions of water for geyser heights for $D_r = 0.055$ m: (a) run C1, at $t = 3.4$ s; (b) run C2, at $t = 3.45$ s; (c) run C3, at $t = 2.15$ s; and (d) run C4, at $t = 1.15$ s.

Table 3. Summary of the performance of orifice plates based on the CFD model

Run series	Orifice size	Maximum geyser height (m) ^a	Maximum pressure at PG1 (kPa)	Maximum pressure in the riser (kPa) ^b	Amount of water flowing out of the riser (L)	Amount of air transported downstream (L)
A2	D_r	0.60	13.9	12.2	3.50	0.32
C0	0	0	19.2	19.2	0	1.83
C1	$0.1D_r$	0.08	16.6	15.9	0.19	1.63
C2	$0.2D_r$	0.11	16.3	105.8	0.94	1.46
C3	$0.4D_r$	0.18	15.3	36.6	1.94	1.13
C4	$0.8D_r$	0.60	13.9	12.5	3.31	0.36

^aMeasured from the top of the riser.

^bMeasured in the riser at 0.76 m above the top of the chamber.

but generates the highest pressure in the upstream pipe. Compared with Run A2, using an orifice plate reduced geyser height by 87%, 82%, 70%, and 0% for Runs C1–C4, respectively. Using an orifice plate reduced the volume of water flowing out of the manhole by 94%, 73%, 45%, and 5.4% for Runs C1–C4, respectively. Using an orifice plate increased the volume of air transported downstream by 5.7, 5.1, 4.5, and 3.5 times the volume for Run A2 in Runs C0–C3, respectively; Run C4 slightly increased this volume. The installation of an orifice plate increased the pressure in the upstream pipe slightly. However, the installation of an orifice plate induced water hammer-like pressure in the riser when the water column reached the orifice plate.

Inflow Control Method for Geyser Mitigation

The D and E runs studied the effect of controlling the flow rate increase with different upstream valve opening times. Two conditions were tested physically. In prototype storm sewer systems, structures such as detention ponds can be installed in order to reduce the increase of the water inflow rate. Considering that the valve was controlled manually, the upstream valve opening time of 0.2 s for Runs D1 and E1 are reasonably referred to as upstream valve opening instantaneous cases.

For Runs D1, D2, and D3, the peak pressure at PG1 dropped by 50% and 79% for Run D2, and D3, respectively, from 47.98 to 23.79 kPa then to 10.01 kPa with the upstream valve opening time (T) of 0.2 s to 10 s and 20 s, respectively. As the upstream valve opening time increased, the rate at which pressure increased in the system was decreased, resulting in decreased peak pressure. No geysers occurred except for mist jetting out of the riser. Therefore, if the upstream valve opening time reaches a certain period with a slower increase in the flow rate, geyser events can potentially be eliminated.

For Runs E1, E2, and E3, with a slow increase of the flow rate in the inlet, the pressure in the upstream pipe increased more slowly, which induced a slow rise of the free surface in the riser and a slow increase of the air flow rate for the air released from the air pocket in the upstream pipe into the chamber. Geysers still occurred but with a lower intensity in terms of geyser height and the amount of water flowing out of the riser. The geysers in these cases were reduced due to the slower increase in the pressure up to final steady state conditions. The final pressures in the system were similar for all E runs, at around 10–12 kPa.

Conclusions

Physical experiments were conducted to investigate the mitigation of geyser events based on an Edmonton manhole that has had frequent geyser issues. A transient 3D numerical model was also implemented to simulate geyser events triggered by a surge front while downstream pipe was almost full and triggered by an entrapped air pocket with an initial water column in the riser. The current setup had a unique connection in which the upstream and downstream pipes had an invert elevation drop of 0.18 m and were connected with the riser by a chamber. Possible ways to mitigate geyser events were assessed. The following conclusions can be drawn based on the Edmonton system configuration:

1. There was no geyser observed in cases with a water recirculation chamber, benching, or an orifice plate for the geyser triggered by a surge front while the downstream pipe was almost full. The WRC induced slight increases in pressure in the system and had a minor effect on the amount of air transported downstream. Benching and orifice plates decreased the pressure in the system and increased the amount of air transported downstream by 2–3 times compared to the system without

retrofitting. The detailed design parameters of the WRC depend on a variety of factors and should be further studied.

2. The installation of an orifice plate at the top of the riser significantly reduced the geyser height and the amount of water ejected out of the riser but dramatically increased the dynamic pressure on the orifice plate—by up to 8 times the original pressure when the orifice opening size was 0.2 times the riser diameter. For the current setup, the installation of an orifice plate also increased the air volume transported to the downstream pipe by more than 5 times the original volume if the orifice diameter was smaller than 0.4 times the riser diameter; this could lead to geyser concerns in downstream areas.
3. The inflow control method reduced geyser events in terms of geyser height and pressure variation. Peak pressure was reduced by as much as 79% when the upstream valve opening time was 30 s when mitigating the geyser triggered by a surge front while the downstream pipe was full.

The current CFD model simulated the air/water interaction during geyser events, but limitations still exist. The current model is unable to model changes in the morphology of water (i.e., from continuous to dispersed phases). Therefore, the actual height of the water droplets during geyser events can only be estimated using the air velocity. Discrepancies are also expected due to the limitations of the geometry and the simplified boundary conditions compared to the prototype structure. The impact of the air transported downstream requires further study. Field monitoring is urgently needed in order to gain a better understanding of the mechanisms of geyser generation in storm water systems and fill the gap between lab-scale and prototype measurements. Also, the scale effect of air/water mixture flow in municipal infrastructures is to be understood.

Data Availability Statement

Some or all data, models, or code generated or used during the study are available from the corresponding author by request, including

1. CFX model setup files;
2. CFX transient result files; and
3. Excel calculations.

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Notation

The following symbols are used in this paper:

- A = cross-sectional area of the riser;
- D = pipe diameter;
- D_r = riser diameter;
- d_0 = diameter of the orifice plate;
- m = mass of the water column in the riser;
- p = pressure;
- Q_w = water inflow rate from the inlet;
- T = upstream valve opening time;
- t = time; and
- v = velocity of the water column in the riser.

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